

EXPERIMENT - 4

TO TEST THE FERROUS IONS IN THE GIVEN AQUEOUS SOLUTION AND OXIDIZE IT TO FERRIC ION, (FERROUS AND FERRIC ION) (REDOX REACTION)

APPARATUS REQUIRED

- | | |
|----------------------|-------------------------|
| a) <u>Beaker</u> | b) <u>Test tube</u> |
| c) <u>Funnel</u> | d) <u>Glass rod</u> |
| e) <u>Wire gauze</u> | f) <u>Bunsen Burner</u> |

CHEMICAL REQUIRED

- | | |
|---|--|
| i. <u>FeSO_4</u> | ii. <u>$\text{Dil. H}_2\text{SO}_4$</u> |
| iii. <u>Conc. HNO_3</u> | iv. <u>$\text{K}_4(\text{Fe}(\text{CN})_6)$</u> |
| v. <u>$\text{K}_3(\text{Fe}(\text{CN})_6)$</u> | vi. <u>$\text{KMnO}_4$</u> |
| vii. <u>NH_4OH</u> | vii. <u>NH_4CNS</u> |

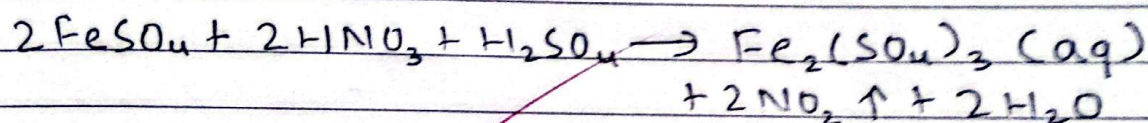
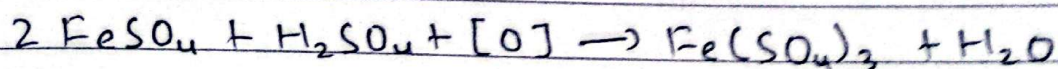
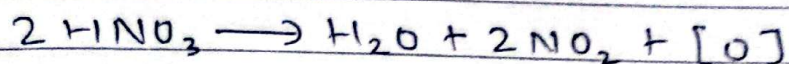
THEORY

The chemical process which involves the loss of electron or increase in oxidation number is oxidation. While the process of gain of electron and decrease in oxidation number is known as reduction.

The residual charge approved on an atom or group of atom when the rest of species are recovered as ions is known as oxidation number.

The redox reaction implies to the special case in which the chemical process of oxidation and reduction occur simultaneously.

A special species which oxidizes the other in a reaction while itself gets reduced is known as the oxidizing agent in a reaction.



PROCEDURE

An amount of 2ml solution was taken in test tube and a few reagents were added to test ferrous ions.

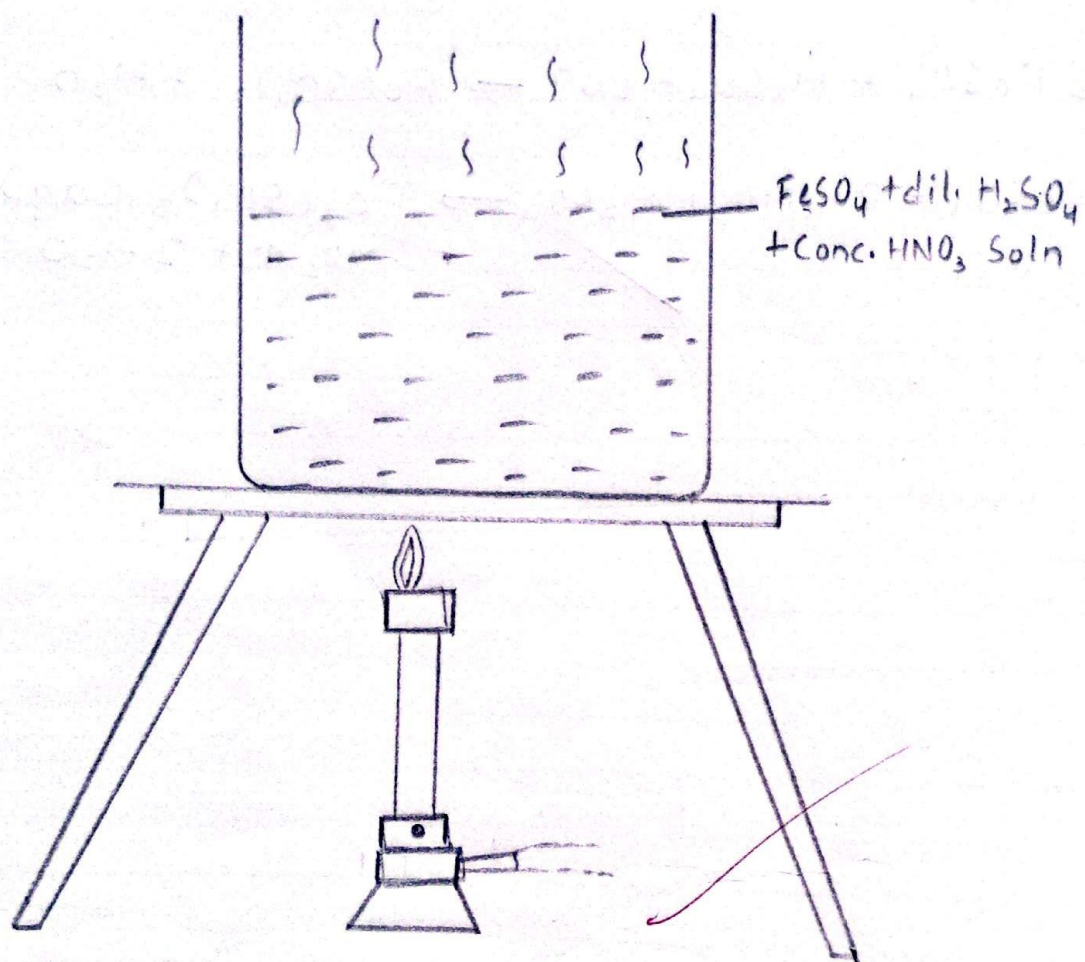


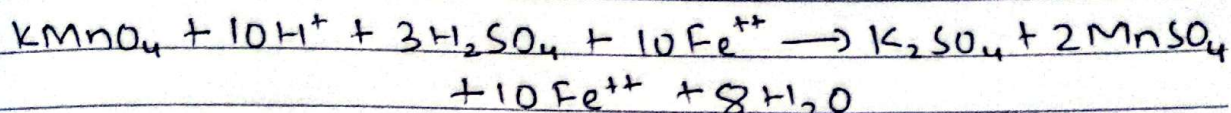
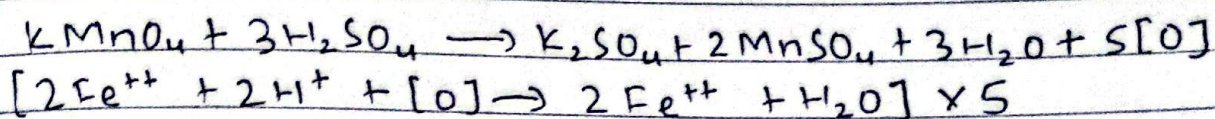
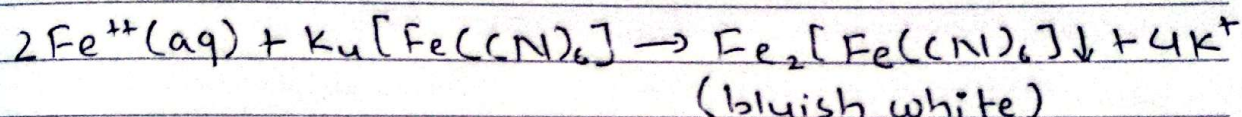
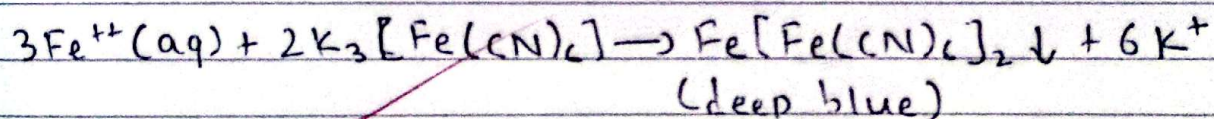
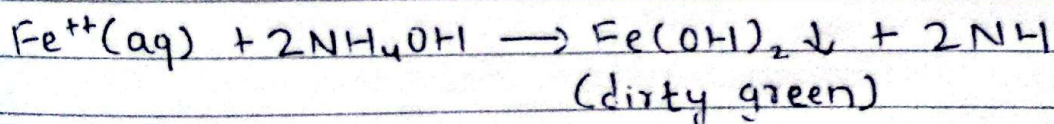
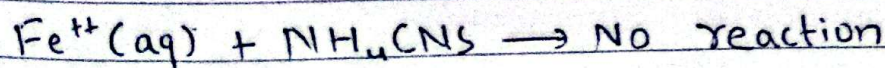
fig-Oxidation of FeSO_4 solution

OBSERVATION TABLE

Test for ferrous ion [Fe^{++}]

S.N.	Experiments	Observations	Inferences
①	1 ml of Original Sol ⁿ was taken in test tube and a few drops of dil. H_2SO_4 and KMnO_4 was added	The pink colour of KMnO_4 discharge	Fe^{++} ion was present in the solution.
②	1 ml of original Sol ⁿ was taken and few drops of $\text{K}_4(\text{Fe}(\text{CN})_6)$ were added	Bluish white ppt was observed	Fe^{++} ion was present in the solution.
③	1 ml of original Sol ⁿ was taken and few drops of NH_4OH were added in test tube	Dirty green ppt was observed.	Fe^{++} ion was present in the solution.
④	1 ml of original Sol ⁿ and Few drops of $\text{K}_3(\text{Fe}(\text{CN})_6)$ were added in test tube	Deep blue ppt was observed	Fe^{++} ion was present in the solution.
⑤	1 ml of original Sol ⁿ and few drops of NH_4CNS sol ⁿ were added	No ppt was observed	Fe^{++} ion was present in the solution.

TEST REACTIONS

① Reaction of Fe^{++} ion with KMnO_4 ② Reaction of Fe^{++} and $\text{K}_4[\text{Fe}(\text{CN})_6]$ ③ Reaction of Fe^{++} and $\text{K}_3[\text{Fe}(\text{CN})_6]$ ④ Reaction of Fe^{++} and NH_4OH ⑤ Reaction of Fe^{++} and NH_4CNS 

OBSERVATION TABLE

Test for ferric ion $[\text{Fe}^{+++}]$

S.N.	Experiments	Observations	Inferences
①	1 ml of Sol^n and few drop of H_2SO_4 and KMnO_4 were added in the test tube	The pink colour was not observed	The sol^n contained Fe^{+++} ion.
②	1 ml oxidized sol^n was taken in the test tube and few drops of $\text{K}_4[\text{Fe}(\text{CN})_6]$ were added	The deep blue colour was observed	Fe^{+++} ion was present in the solution
③	1 ml of oxidized Sol^n and Few drops of $\text{K}_3[\text{Fe}(\text{CN})_6]$ were added.	Ppt and green coloration was not appeared.	Fe^{+++} ion was present in the solution.
④	1 ml of Oxidized Sol^n and Few drops of NH_4OH Sol^n were added in the test tube.	The brown ppt was observed.	Fe^{+++} ion was present in the solution.
⑤	1 ml of Oxidized Sol^n was taken in the test tube and Few drops of NH_4CNS sol^n .	The blood red colour was observed.	Fe^{+++} ion was present.

TEST REACTIONS

① Reaction of Fe^{+++} and acidified KMnO_4
 $\text{KMnO}_4 + \text{H}_2\text{SO}_4 + \text{Fe}^{+++} \rightarrow \text{No reaction}$

② Reaction of Fe^{+++} and $\text{K}_4[\text{Fe}(\text{CN})_6]$
 $4\text{Fe}^{+++} + 3\text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow \text{Fe}[\text{Fe}(\text{CN})_6]_3 + 12\text{K}^+$
 (deep blue)

③ Reaction of Fe^{+++} ion and NH_4OH
 $\text{Fe}^{+++} + 3\text{NH}_4\text{OH} \rightarrow \text{Fe}(\text{OH})_3 + 3\text{NH}_4^+$
 (green colour)

④ Reaction of Fe^{+++} ion and $\text{K}_3[\text{Fe}(\text{CN})_6]$
 $\text{Fe}^{+++} + \text{K}_3[\text{Fe}(\text{CN})_6] \rightarrow \text{Fe}[\text{Fe}(\text{CN})_6] + 3\text{K}^+$
 (Brown ppt)

⑤ Reaction of Fe^{+++} and NH_4CNS
 $\text{Fe}^{+++} + 3\text{NH}_4\text{CNS} \rightarrow \text{Fe}(\text{CNS})_3 \downarrow + 3\text{NH}_4^+$
 (Blood red)

RESULT

So, after the oxidation of ferrous ion to Ferric ion, different test were carried out and test results were submitted for inspection.

CONCLUSION

Hence, From above experiment it is concluded that ion present in sample can be oxidized and both of the ions. i.e. Oxidized and non-oxidized can be detected by carrying out various tests.

PRECAUTIONS

- ① Personal protective equipment should be worn.
- ② chemical and glass wares should be handled carefully.

VIVA-VOCE

1. What are chemical names of Fe^{++} and Fe^{+++} ?
2. What is oxidation number of ferrous and ferric ions?
3. What is oxidation number of Fe in following compounds a) $\text{K}_3[\text{Fe}(\text{CN})_6]$ and b) $\text{K}_4[\text{Fe}(\text{CN})_6]$?
4. Why is reaction between acidified solution FeSO_4 (Acidified with dil. H_2SO_4) with conc. HNO_3 is called redox reaction?
5. What are color of $\text{Fe}^{++}(\text{aq.})$ and $\text{Fe}^{+++}(\text{aq.})$?

① Ferrous and Ferric ion respectively.

② The oxidation number of ferrous ion is +2 and Ferric ion is +3

③ a. Fe is in +3 oxidation state
b. Fe is in +2 oxidation state

④ This is because in this reaction oxidation number of Fe in FeSO_4 gets changed from +2 to +3 and oxidation number of N in HNO_3 changes from +5 to +4. i.e FeSO_4 gets oxidized and HNO_3 gets reduced Hence, it is called redox reaction.

⑤ Colour of $\text{Fe}^{++}(\text{aq.})$ is light green and $\text{Fe}^{+++}(\text{aq.})$ is yellowish brown.